**Information Assurance and Security**

**WEEK 11 Coursework**

**This is a coursework for a group of 3 students. Individual work in not allowed. Perform the activity as directed and provide a screenshot of each activity as you perform it.**

## ZoneAlarm Free Firewall

ZoneAlarm is a PC-based firewall and IDPS tool. Visit the product manufacturer at www.zonelabs.com, and find the product specification for the IDPS features of ZoneAlarm. Which of the ZoneAlarm products offer these features? (20 pts)

### Core Firewall Protection

- **Two-Way Firewall**: Monitors both incoming and outgoing network traffic to block potential threats, prevent unauthorized access, and stop spyware from sending your data out to the internet.
- **Invisibility Mode**: Makes your PC hidden from hackers on the network.
- **Traffic Monitoring**: Meticulously monitors data packets entering and exiting your computer.

### Advanced Security Features

- **Advanced Anti-Bot Safeguard**: Actively detects and blocks harmful bots in real-time, preventing botnet infiltration, spamming, data theft, and access to suspicious websites.
- **WiFi PC Protection**: Monitors programs for suspicious behavior and protects you from hackers and identity thieves when connected to unsecured networks (like public WiFi).
- **Advanced-Access Protection**: Targets and blocks advanced attacks that other firewalls might miss, including raw data access, timing attacks, and SCM & COM-based attacks.
- **Early Boot Protection**: Detects whether your PC is running a trusted operating system or a dangerous rootkit, protecting your computer during the startup process.
- **Component Control**: Protects against malware that tries to bypass the firewall by loading malicious DLLs into a trusted application.

### Usability & Control Features

- **Advanced Settings Control**: Allows you to classify networks as "private zone" (home) or "public zone" (untrusted) to automatically increase security and reduce attack vectors based on your location.
- **Program Access Control**: Lets you block unwanted traffic and control which programs can access the internet.

**System & Operational Details**

- **Compatibility**: Fully compatible with Windows 11 and Windows 10 (32-bit and 64-bit).
- **Performance Impact**: Designed to minimize effect on system performance and internet connection speed.
- **Automatic Updates**: Regularly checks for and installs updates (new threat definitions, security rules, features) automatically without manual intervention.
- **Default Security**: Installer is configured with default settings that ensure optimal security, requiring no additional adjustments.

**Limitations**

- **Not Complete Protection**: The free version may not provide complete protection against all types of cyber threats.
- **Compatibility Restriction**: ZoneAlarm is only compatible with MS Windows Defender and is not compatible with any other anti-malware software. You must uninstall other anti-malware software before installing ZoneAlarm to avoid system stability issues.

**Support & Requirements**

- **Support Options**: Access to online support center, live chat, and email assistance.
- **System Requirements**:
    - OS: Windows 10 or 11
    - RAM: 3 GB (32-bit) or 4 GB (64-bit)
    - Processor: 2 GHz or faster
    - Hard Disk Space: 1.5 GB available
    - Microsoft .NET Framework version 3.5 or later (will be installed automatically if missing)

| **Feature / Capability** | **ZoneAlarm Free Firewall** | **ZoneAlarm Pro Antivirus + Firewall** | **ZoneAlarm Extreme Security NextGen** |
|---|---|---|---|
| Two-Way Firewall & Stealth Mode | ✓ | ✓ | ✓ |
| SmartDefense Advisor (Auto-Pilot) | ✓ | ✓ | ✓ |
| Basic Malicious URL Blocking | ✓ | ✓ | ✓ |
| OSFirewall (Behavioral IDPS) | X | ✓ | ✓ |
| Advanced Download Protection | X | ✓ | ✓ |
| Antivirus / Anti-Spyware Engine | X | ✓ | ✓ |
| Anti-Keylogger / Privacy | X | ✓ | ✓ |
| Threat Emulation (Cloud Sandbox) | X | X | ✓ |
| Threat Extraction (Safe Docs) | X | X | ✓ |
| Anti-Ransomware (File Restore) | X | X | ✓ |
| Mobile OS Support | X | X | ✓ |

2. Using the Internet, search for commercial IDPS systems. What classification systems and descriptions are used, and how can these be used to compare the features and components of each IDPS? Create a comparison spreadsheet identifying the classification systems you find. (20 pts)

- Commercial Intrusion Detection and Prevention Systems (IDPS) are primarily classified by the environment they monitor into four main categories: Network-Based (NIDPS) which monitors traffic across network segments; Host-Based (HIDPS) which monitors the internal activity and files of a single specific device; Wireless (WIDPS) which analyzes wireless network protocols and radio frequencies; and Network Behavior Analysis (NBA) which examines overall traffic flows for unusual patterns. Understanding these classifications allows organizations to easily compare features across different vendors. For example, by knowing a system is an NIDPS, a buyer knows to compare its packet-processing speed and inline sensor components, whereas comparing HIDPS systems involves looking at endpoint agent performance, file integrity monitoring, and CPU impact on the host.

| Vendor / Commercial IDPS | Primary Classification | Primary Classification | Key Components & Features |
|---|---|---|---|
| **Cisco Secure IPS** (formerly Firepower) | **NIDPS** (Network-Based) | Signature & Stateful Protocol Analysis | Inline network sensors, centralized management console, deep packet inspection, NGFW integration. |
| **Trellix Endpoint Security** (formerly McAfee/FireEye) | **HIDPS** (Host-Based) | Anomaly & Signature | Endpoint agents, file integrity monitoring, system log analysis, ransomware rollback. |
| Darktrace /Network | **NBA** (Network Behavior Analysis) | AI / Anomaly-Based | Self-learning AI engines, traffic flow sensors, real-time topology visualization. |
| Palo Alto Threat Prevention | **NIDPS** (Network-Based) | Signature & Anomaly | Single-pass architecture, automated threat intelligence cloud updates, inline blocking. |

| | | | |
|---|---|---|---|
| Aruba (HPE) AirWave | **WIDPS** (Wireless) | Signature & Protocol Analysis | Wireless access point sensors, RF spectrum monitoring, rogue AP containment. |
| CrowdStrike Falcon Insight | **HIDPS** (Host-Based) | Behavior / Anomaly | Lightweight cloud-managed agents, endpoint detection and response (EDR), threat hunting. |

3. Use the Internet to find vendors of thumbprint and iris scanning tools. Which of these tools is more economical? Which of these is least intrusive? (20 pts)

| **Technology** | **Vendor** | **Product Example** | **Key Feature** |
|---|---|---|---|
| **Thumbprint/Fingerprint** | Miaxis Biometrics Co. | SM-91M | FBI-certified, compact, portable scanner |
| | Union Biometrics | NScan-T | FBI-certified, captures single or four fingers, rugged (IP54) |
| | IDEMIA | TouchPrint Enterprise (TPE) LiveScan | Multi-modal (fingerprint, iris, face), for law enforcement |
| **Iris** | IriTech, Inc. | IriAegis Series (BK, 26T Pro, FI25) | Contactless, fast capture, free SDK, models for mobile or access control |
| | BioID Technologies Ltd. | BioIRIS | Dual-eye capture, compact, meets ISO standards |

**Which is more economical?**

- Fingerprint scanning technology is generally more economical than iris scanning. This is largely because fingerprint technology has been adopted more widely and for a longer period, leading to economies of scale that make the hardware less expensive.

**Which is less intrusive?**

- In terms of physical contact, iris scanning is less intrusive. These systems are contactless and capture an image from a distance (some models up to 80cm away), which many users find more hygienic and convenient. Fingerprint scanners, by contrast, typically require physical contact with a sensor, which some may perceive as intrusive or unsanitary but in terms of personal privacy, fingerprint scanning is often seen as less intrusive. Some people feel that scanning their iris, an image of their eye, is a greater invasion of personal privacy than using a fingerprint.

However, it's worth noting that the iris, unlike a face, cannot be easily identified by a human operator without the help of specialized technology, which can be a privacy advantage in some contexts.

4. There are several online passphrase generators available. Locate at least two of them on the Internet, and try them out. What did you observe? (20 pts)

We used Proton and Keepersecurity passphrase this are free passphrase generators. This generator creates a password randomizer and creates a secure that is hard to crack or brute force attack. The generator give us a selection on how many word with the combination of capital letters, numbers and separators like hyphens, spaces, period, commas, underscores, numbers and numbers/symbols and other things this creates a combination that is hard to bypass or brute force we use passwordstrength a website which can estimate the password strength of the password this shows how the password generated is that hard to be cracked

5. What were your significant learnings on this coursework? (20pts)

We learned that layered security is needed to the extra security since firewalls alone cannot provide complete protection and must be complemented by Intrusion Detection and Prevention System (IDPS) tools that add behavioral and anomaly detection. We also discovered that understanding classification systems such as Network IDPS, Host-based IDPS, Wireless IDPS, and Network Behavior Analysis is critical for evaluating and comparing commercial intrusion detection and prevention solutions. We also realized that fingerprint technology is generally more economical, while iris scanning is less intrusive because it is contactless, though perceptions of privacy vary. When working with passphrase generators We observed that length and randomness are sure more important than simple complexity tricks like adding numbers at the end.

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